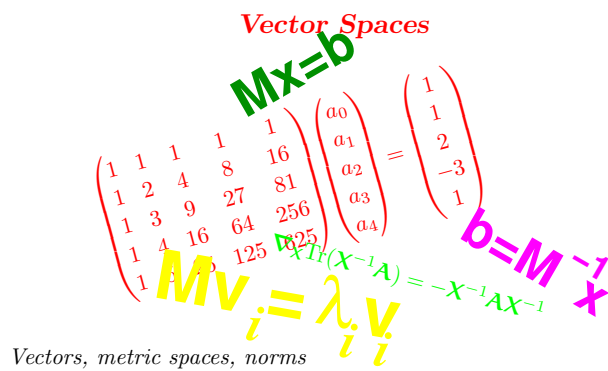




## Matrices, Vectors and All That

- The language of machine learning is mathematics
- Sometimes we draw pretty pictures to explain the mathematics
- Much of the mathematics we will use involves vectors, matrices and functions
- You need to master the language of mathematics, otherwise you won't understand the algorithms
- I'm going to spend this lecture and the next revising the mathematics you need to know (but I'm going use a slightly posher language than you are probably used to)

## Vectors

- We often work with objects with many components (features)
- To help handle this we will use vector notation
  - We represent vectors by bold symbols
  - All our vectors are column vectors by default
  - We treat them as  $n \times 1$  matrix
- We write row vectors as transposes of column vectors

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$\mathbf{y}^T = (y_1, y_2, \dots, y_n)$$

## Vector Space

- A vector space,  $\mathcal{V}$ , is a set of vectors which satisfies
  1. if  $\mathbf{v}, \mathbf{w} \in \mathcal{V}$  then  $a\mathbf{v} \in \mathcal{V}$  and  $\mathbf{v} + \mathbf{w} \in \mathcal{V}$  (closure)
  2.  $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$  (commutativity of addition)
  3.  $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$  (associativity of addition)
  4.  $\mathbf{v} + \mathbf{0} = \mathbf{v}$  (existence of additive identity  $\mathbf{0}$ )
  5.  $1\mathbf{v} = \mathbf{v}$  (existence of multiplicative identity 1)
  6.  $a(b\mathbf{v}) = (ab)\mathbf{v}$  (distributive properties)
  7.  $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$
  8.  $(a+b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

(You don't need to remember these)
- Just from these properties we can deduce other properties



1. **Vector Spaces**
2. **Metrics (distances)**
3. **Norms**

## Scalars (Fields)

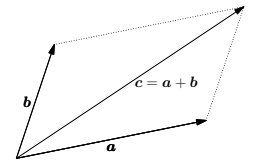
- Vector spaces involve **fields** (numbers)—aka **scalars**
- These are quantities we can add together ( $a + b$ ) and multiply together ( $a \times b$ )
- Formally they form an Abelian group under addition with an identity 0 and excluding 0 an Abelian group under multiplication and they are distributive

$$a \times (b + c) = a \times b + a \times c$$

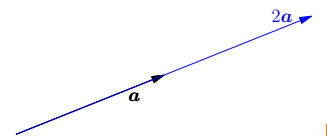
- Although this sounds rather daunting don't panic. They behave like numbers. The field might be integers, rational numbers, reals, complex numbers or something a bit more exotic—but we will almost always consider reals

## Basic Vector Operations

- The basic vector operations are adding

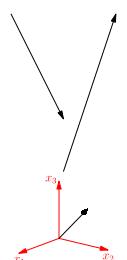


- multiplying by a scalar (a number)



$$\mathbb{R}^n$$

- When we first learn about vectors we think of them as arrows in 3-D space
- If we centre them all at the origin then there is a one-to-one correspondence between vectors and points in space
- We call this vector space  $\mathbb{R}^3$
- Any set of quantities  $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$  which satisfy the axioms above form a vector space  $\mathbb{R}^n$
- Of course, we can't so easily draw pictures of high-dimensional vectors



- Any set of object that satisfies the axioms of a vector spacer are **vectors**—not just  $v \in \mathbb{R}^n$
- Matrices satisfy all the conditions of a vector space
- Infinite sequences form a vector space
- Functions form a vector space
  - Let  $C(a,b)$  be the set of functions defined on the interval  $[a,b]$
  - Note that if  $f(x), g(x) \in C(a,b)$  then  $af(x) \in C(a,b)$  and  $f(x) + g(x) \in C(a,b)$
- Bounded vectors in  $\mathbb{R}^n$  **don't** form a vector space

## Metrics

- Vector spaces become more interesting if we have a notion of distance
- We say  $d(x,y)$  is a **distance** or **metric** if
  - $d(x,y) \geq 0$  (non-negativity)
  - $d(x,y) = 0$  iff  $x = y$  (identity of indiscernibles)
  - $d(x,y) = d(y,x)$  (symmetry)
  - $d(x,y) \leq d(x,z) + d(z,y)$  (triangular inequality)
- There are typically many possible distances (e.g. Euclidean distance, Manhattan distance, etc.)
- Often one or more condition isn't satisfied then we have a **pseudo-metric**

## Lipschitz Function

- One way to characterise a well behaved function,  $f(x)$ , is if there exists a number  $K < \infty$  such that for all  $x$  and  $y$ 

$$d(f(x), f(y)) \leq K d(x, y)$$
- This is known as a **Lipschitz condition** and the function is said to be  $K$ -Lipschitz or Lipschitz continuous
- Note that such functions cannot have any jumps (i.e. they are continuous)
- The size of  $K$  measures the limit on the amplifying effect of the function

## Contractive Mappings

- An interesting class of function are those for which  $K < 1$
- These are said to be contractive mappings
- A famous theorem that applies to contractive mappings is the Banach fixed-point theorem which says there exists a unique fixed point such that  $f(x) = x$
- This is used for example in showing that various algorithms will converge

- Vector Spaces
- Metrics (distances)**
- Norms

$$Mx=b$$

$$Mv_i = \lambda_i v_i$$

$$b=M$$

## Mappings and Functions

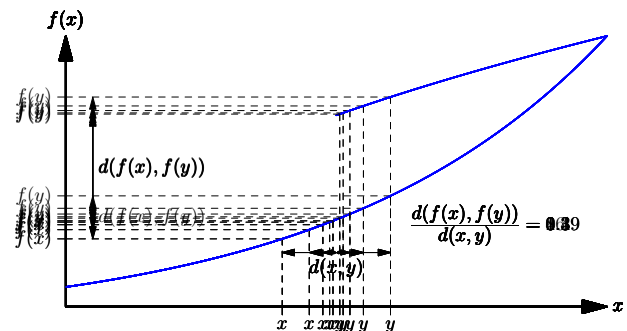
- A function defines a mapping from one vector space to another (although the spaces might be the same) e.g.

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

( $f$  maps the reals onto reals, i.e.  $f(x)$  takes a real  $x$  and gives you a new real number  $y = f(x)$ )

- We are often interested in functions that behave nicely
- E.g. They are continuous

## Lipschitz Continuous Functions



- Definition is applicable to any function defined in a metric space

## Outline

- Vector Spaces
- Metrics (distances)**
- Norms**

$$Mx=b$$

$$Mv_i = \lambda_i v_i$$

$$b=M$$

## Norms

- Vector spaces are even more interesting with a notion of length
- **Norms** provide some measure of the size of a vector
- To formalise this we define the **norm** of an object  $v$  as  $\|v\|$  satisfying
  1.  $\|v\| > 0$  if  $v \neq 0$  and  $\|0\| = 0$  (non-negativity)
  2.  $\|av\| = a\|v\|$  (“linearity”)
  3.  $\|u + v\| \leq \|u\| + \|v\|$  (triangular inequality)
- When some criteria aren’t satisfied we have a **pseudo-norms**
- Norms provide a metric  $d(x, y) = \|x - y\|$  (they are metric spaces)

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## Matrix Norms

- We can define norms for other objects
- The norm of a matrix encodes how large the mapping is
- The Frobenius norm is defined by

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |A_{ij}|^2}$$

- Many other norms exist including 1-norm, max-norm, etc.
- Some, but not all, norms satisfy the inequality

$$\|AB\| \leq \|A\| \times \|B\|$$

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## Why Should You Care?

- Deep learning involves multiplying the input (which we can think of as a vector  $x$ ) by many layers
- In CNNs we have convolutional layers and dense layers
- The effect of applying these layers can be represented by a matrix multiplication  $x_n = L_n x_{n-1}$
- We also do other things like applying ReLU’s or pooling that changes the magnitude,  $x_n$ , of our representation
- If you are developing new architectures you want  $\|x_n\|$  neither to blow up or vanish
- This can be controlled by carefully choosing  $\|L_n\|$

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## Norms and Metrics

- Given a norm, there exists an induced distance
$$d(x, y) = \|x - y\|$$
- Any normed space has an associated metric space
- Thus we can define the Frobenious-distance between matrices

$$d_F(A, B) = \|A - B\|_F$$

- It is easy to see that non-negativity, linearity and triangular inequality for distances are all inherited from the same property for norms

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## Vector Norms

- The familiar vector norm is the (Euclidean) two norm

$$\|v\|_2 = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

- Other norms exist, such as the  $p$ -norm ( $p \geq 1$ )

$$\|v\|_p = \left( \sum_{i=1}^n |v_i|^p \right)^{1/p}$$

- Special cases include the 1-norm and the infinite norm

$$\|v\|_1 = \sum_{i=1}^n |v_i| \quad \|v\|_\infty = \max_i |v_i|$$

- The 0-norm is a pseudo-norm as it does not satisfy condition 2

$$\|v\|_0 = \text{number of non-zero components}$$

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## Compatible Norms

- A vector and matrix norm are said to be compatible if

$$\|Mv\|_b \leq \|M\|_a \times \|v\|_b$$

(Spectral and Euclidean norms are compatible)

- Norms provide quick ways to bound the maximum growth of a vector under a mapping induced by the matrix
- We will see that a measure of the sensitivity of a mapping is in terms of the ratio of its maximum effect to its minimum effect on a vector
- This is known as the **conditioning**, given by  $\|M\| \times \|M^{-1}\|$

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## Function Norms

- Functions can also have norms, for example, if  $f(x)$  is defined in some interval  $\mathcal{I}$

$$\|f\|_{L^2} = \sqrt{\int_{x \in \mathcal{I}} f^2(x) dx}$$

- The  $L^2$  vector space is the set of function where  $\|f\|_{L^2} < \infty$
- The  $L^1$ -norm is given by  $\|f\|_{L^1} = \int_{x \in \mathcal{I}} |f(x)| dx$
- The infinite-norm is given by  $\|f\|_\infty = \max_{x \in \mathcal{I}} |f(x)|$

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## Summary

- Vector spaces with a distance (metric spaces) and vector spaces with a norm (normed vector spaces) are interesting objects
- They allow you to define a topology (open/closed sets, etc.)
- You can build up ideas about connectedness, continuity, contractive maps, fixed-point theorems, . . .
- For the most part we are going to consider an even more powerful vector space that has an inner-product defined

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